A. Course: CSCI 414/514 - Platform Based Development
Group Members: Chidubem Okoye, Yvonne Onmakpo, Tonika Devi Avanigadda

Project Title: HuskyPark Predictor: AI Driven Parking Spot Probability Dashboard for SCSU

B. Project Idea description

What Problem Are You Addressing?

Parking on campus could be a hassle. We will be working on making it less stressful using St. Cloud State University as our patient zero. At St. Cloud State University (SCSU), students, faculty, and visitors frequently face challenges finding available parking spots due to high demand, varying lot restrictions, and unpredictable factors like class schedules, events, and Minnesota weather conditions. With over 24 surface lots, 8 pay lots, and the 4th Avenue Parking Ramp, parking is fragmented across categories: resident lots (e.g., RES in 3 locations, N Lot), commuter/student lots (e.g., A, C, E, K, M, V, Stateview), employee lots (e.g., AA, B, D, F, G, H, I, J, L, O, P, R, S, W), and visitor pay lots (e.g., Eastman, Husky, ISELF, Miller, South). Permits are required in surface lots at all times, displayed from the rearview mirror, and cost around $192 for a full spring semester surface lot permit (prorated based on purchase date). This leads to time wasted circling lots, increased stress, higher emissions from idling vehicles, and potential late arrivals to classes or meetings.
Our project addresses this by creating a web-based dashboard that predicts the probability of finding an available spot in specific lots or zones, based on historical data, real-time crowdsourced reports, and external factors like the time of day, day of the week, campus events, and weather.

Why Is It Important or Interesting?

Parking inefficiencies are a daily frustration for the SCSU community, especially in a midsized campus where spots are limited and demand peaks during class hours (e.g., mornings and afternoons). It's important because it promotes efficiency, sustainability (reducing unnecessary driving), and accessibility; helping commuters save time and residents avoid fines (e.g., parking in restricted areas). It's interesting as it applies course concepts like multi database integration for data storage, generative AI for user friendly recommendations, and DevOps for a reliable, scalable deployment. Plus, as a local solution tailored to SCSU's layout along the Mississippi River, it has real potential for campus adoption.

Who Would Benefit Your Solution?

Students (Residents and Commuters): Quick predictions for lots like RES, N, C, M, V, or K, reducing hunt time and integrating with class schedules.
Faculty/Staff (Employees): Insights for employee-specific lots like AA, B, D, etc., aiding in planning around meetings.
Visitors/Guests: Guidance to pay lots like Husky or 4th Avenue Ramp, with probability estimates to avoid full lots.
University Administration: Aggregated data on usage patterns for better management and future planning.
Broader Community: Reduced traffic congestion around campus in Saint Cloud, MN.

C. Motivation

Our group chose the HuskyPark Predictor project because parking availability is a recurring and personally relevant issue experienced by students, faculty, and visitors at St. Cloud State University. As daily campus commuters, we have observed firsthand how unpredictable parking conditions lead to wasted time, increased stress, and occasional tardiness to academic and professional commitments. This real-world relevance makes the project both meaningful and impactful, as it addresses a practical challenge that affects a large portion of the SCSU community.

In addition, the project aligns closely with current trends in smart campus solutions and data-driven decision-making. Universities are increasingly adopting intelligent systems to improve mobility, sustainability, and user experience. By developing a predictive parking probability dashboard tailored specifically to SCSU, our project contributes to the broader concept of smart campus infrastructure while remaining grounded in a localized, realistic use case.

The idea is also academically motivating because it allows our group to apply platform-based development concepts such as data integration, analytics, dashboard visualization, and AI-assisted recommendations within a single cohesive system. Rather than creating a purely theoretical application, we are designing a solution that has direct practical value and potential for future campus adoption.

Feasibility

The HuskyPark Predictor project is feasible within the scope and timeline of this course because it focuses on a web-based analytical platform rather than physical hardware implementation. Instead of requiring real-time sensor installation in parking lots, the system will utilize simulated datasets, historical parking patterns, crowdsourced inputs, and publicly available contextual data such as time schedules, weather conditions, and campus events to generate probability-based predictions.

From a technical perspective, the project can be realistically implemented using technologies that are appropriate for platform-based development. The backend will support data storage and processing, while the frontend dashboard will provide visual insights and user interaction. The predictive component can be implemented using basic machine learning or statistical modeling techniques, which are achievable at the undergraduate level without requiring overly complex AI infrastructure.

The modular design of the system enhances feasibility. The project can be divided into manageable components, including data collection, database management, predictive modeling, and dashboard visualization. This structure allows team members to work in parallel and ensures steady progress throughout the development cycle.

Our group also plans to use accessible and well-documented tools such as web development frameworks, database systems, and version control platforms. These technologies are widely supported, scalable, and suitable for academic projects, making the implementation both realistic and achievable within the semester timeframe. The project is challenging yet practical, aligns with the course objectives in platform-based system design, and can be successfully completed using existing technical skills and structured team collaboration.

INNOVATION / UNIQUENESS:

HuskyPark Predictor is unique because it goes beyond a basic “parking status” app and instead delivers a probability-based prediction dashboard tailored to SCSU’s real parking structure (resident, commuter, employee, and pay lots). Rather than showing static availability, the platform estimates the likelihood of finding a spot in specific lots or zones, helping users choose where to go first and reducing time wasted driving around campus.

The project also stands out by using a hybrid data approach-combining historical/simulated patterns, crowdsourced reports, and contextual factors such as time of day, day of week, campus events, and weather. This supports course objectives by demonstrating a full-stack platform with database-driven analytics and the option to add AI-generated explanations or recommendations to make results more user-friendly.

EXPECTED OUTCOME:

By the end of the project, we will deliver a working web-based dashboard where users can select a permit category (student, resident, employee, visitor) and view parking probability indicators for selected SCSU lots. Users will be able to submit quick lot-status reports (e.g., found

spot/full/hard to find), and the system will update the dashboard based on recent inputs and time-based patterns.

Technically, the platform will include a backend API, a database to store lots and reports, and a probability-scoring component that updates predictions in near real time. The system will be containerized for easy deployment, with clear documentation and a basic plan for testing and evaluation using simulated data and collected reports.

TABLE OF CONTRIBUTION

STUDENT NAMES	CONTRIBUTION/ EXPECTED ROLE	PERCENTAGE
CHIDUBEM OKOYE	Planning & documentation, backend integration	33.33 %
Yvonne Onmakpo	Data modeling & probability logic, testing	33.33 %
Tonika Devi Avanigadda	UI mockups, visualization & frontend implementation	33.34 %